

Department of Mechatronics and Industrial Engineering			
Odd semester Examination – Academic year 2023/2024			
Subject	INTRODUCTION TO ENGINEERING SCIENCE		
Class	B4MI/E	Date	04 / 01 / 2024
Max. Marks	60	Time	2 hours
Student Name	[REDACTED]	Class	B4MI
		Reg. No	[REDACTED]

Question No.	Q1	Q2	Q3	Q4	Q5	Q6	Total
Maximum Marks	21	13	13	13	13	13	TOTAL Out of 60
Marks Scored							
Lecturer	Dr. Fawaz Ahmed Ghaleb					Signature	

NOTES: Answer Question No. (1) and any other **THREE** questions;

QUESTION (1)

Fill the following statement

- Engineers play a significant role in the design and maintenance of a nation's infrastructure, including Communication systems, Public Utilities, and transportation.
- Engineers consider important factors such as cost, efficiency, reliability, and safety when designing these products.
- Good engineers have time-management skills that enable them to work productively and efficiently.
- Engineering Science is a broad discipline that encompasses many different scientific principles and associated mathematics that underlie engineering.
- Mechatronics & and industrial engineering developed in direct response to industrial demand for engineers with multi-disciplinary skills is a combination of mechanical, electronics, control, computer and systems design engineering streams.
- Engineers are problem solvers. Once they have obtained a solution to a problem, they need to communicate effectively their solution to various people inside or outside their organization.
- The title of a report should be brief informative description of the report content.
- The study of ethics helps students to develop widely applicable skills in communication, reasoning and reflection.
- Moral values and responsible conduct will play a crucial role in decision making.
- Engineers must perform under a standard of professional behavior which requires adherence to the highest principles of ethical conduct.

QUESTION (2)

Explain briefly why?

1. Why is mechatronics & Industrial Engineering department being popular?

Because of the wide and equal opportunities work in the other areas such as electrical, system, computer engineering, etc.

2. Why is basic science important in engineering?

For understanding how the world works and developing new technologies.

3. Why is an abstract important to the readers?

Because readers often read this part first to decide if they should read the report or not.

4. Why are good oral and written communication skills important in engineers?

Because communicate your solution to a problem is as important as the solution itself. Cause the results of all your effort may not be understood and appreciated if you cannot communicate effectively.

5. Why is technical report important in engineering?

Because it's dealing with experimental investigations.

QUESTION (3)

Underline the correct word (S) in the following:

1. is the final step of any engineering analysis.

(a). Defining the Problem

(b). Verifying the Results

(c). Simplifying the problem

2. should be a brief informative description of the report contents.

(a). Conclusions and Recommendations

(b). Title

(c). Abstract

3. used to transforming science into useful products for human comfort.

(a). Science

(b). Engineering

(c). Engineering Science

4. are defined as the comprises the fundamental principles of natural sciences such as physics, chemistry, and biology.

(a). Marine Engineering

(b). Basic sciences

(c). Architecture Engineering

5. The purpose of the is to state what is to be investigated through the performance of the experiment.

(a). Theory and Analysis

(b). Objectives section

(c). Appendix

QUESTION (4)

a) What do you understand by professional ethics?

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b) Differentiate between engineering communication and engineering ethics?

Engineering Communication	Engineering Ethics

QUESTION (5)

Which of the following statement are false? Correct them.

1. Science is transforming science into useful products for human comfort.

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2. The major branches within the Mechanical engineer's discipline include structural, environmental, transportation, water resources, and geotechnical

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3. Computer Science Engineering (CSE) is the technology required for information processing

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4. Defining the Problem is the final step of any engineering analysis should be the verification of results.

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5. The purpose of the appendix section is to state what is to be investigated through the performance of the experiment.

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6. The purpose of the objectives section is to provide the reader with copies of all original data sheets, diagrams, and supplementary notes.

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7. The conclusions and recommendations section compare your objectives with your experimental results.

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QUESTION (6)**Match part I and II**

I							II			
1-	It is the combination between Engineering and Science.						A	Civil Engineering		
2-	To provide the reader with copies of all original data sheets, diagrams, and supplementary notes						B	Abstract		
3-	It is the ability as well as responsibility of an engineer to judge his decisions from the context of the general wellbeing of the society						C	Executive summaries		
4-	It is a practitioner belonging to a specific profession.						D	Electrical Engineering		
5-	It is very important that all information be conveyed in a manner easily understood by the listener.						E	Oral Communication and Presentation		
6-	It is a very important part of a report						F	A professional Ethics		
7-	Are means of communicating to people in top management positions, such as a vice president of a company, the findings of a detailed study or a proposal.						G	Engineering Science		
8-	It is including structural, environmental, transportation, water resources, and geotechnical.						H	Appendix		
9-	It is including power generation, power transmission and distribution, and controls.						I	Engineering Science		
10-	It is a broad discipline that encompasses many different scientific principles and associated mathematics that underlie engineering.						J	Ethics in engineering		
I	1	2	3	4	5	6	7	8	9	10
II	I	H	J	F	E	B	C	A	D	G